



Samson Niyizurugero

Solution Engineer | Full-Stack Developer | Payments, Integrations and Secure Systems

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Professional Profile

Solution Engineer and Full-Stack Developer with an MSc in Mathematical Sciences from AIMS Rwanda (2023) and over 3 years of experience delivering production business applications in financial services and digital health. Currently a Full-Stack Developer Fellow at Kosmotive through the AIMS-H2i Data Science Fellowship (Health Entrepreneurship, Africa Health Collaborative, University of Toronto), building platforms that reach Rwanda, DRC, Central Africa and Kenya. Experienced across the full lifecycle: design, development, third-party integration, automated testing, database engineering, security controls, CI/CD, and 2nd/3rd-line production support. Strong background in regulated environments, audit trails, data governance, and OWASP-aligned secure coding.

Flagship Platforms: Kosmotive Ecosystem (KosmoPads, KosmoHealth, PayGo)

Digital lending, health-commerce, and flexible-payment platforms serving reproductive, maternal and child health access across Rwanda, DRC, Central Africa and Kenya. Stack: Node.js, Express, TypeScript, React, PostgreSQL (Supabase), Drizzle ORM, Zod, Tailwind, GitHub Actions, Render. Integrations: MTN MoMo Collections API, Resend email, SMS gateway via REST with signed webhooks.

Audit-Readiness Controls Implemented and Verified

- Full-stack architecture with strict TypeScript typing; PR templates and CODEOWNERS enforcing peer review before merge to main.
- README, architecture overview, database schema, API reference and data-flow diagrams versioned inside the repo under `/docs`.
- GitHub Actions CI/CD running install, build, lint and automated tests on every push and PR, with deployment blocked on failure.
- Jest and Vitest suites covering the repayment and penalty engine and payment reconciliation first; coverage reported per run.
- Formal issue tracker linking each defect to its root cause, fix commit and verification evidence.
- MTN MoMo webhook handler with structured failure logging, atomic idempotent confirmation, and a persistent retry queue with exponential backoff.
- Asynchronous processing via a Redis/BullMQ-style queue for webhook retries and background jobs, with the architecture decision documented.
- Relational schema of 18 normalised tables, 29 foreign keys and 8 unique constraints, with `drizzle-kit` migrations checked in.
- Secondary indexes on loan status, customer lookup, order and payment status for high-frequency admin and reporting queries.
- Application-layer logic via Drizzle ORM documented as a deliberate, versionable alternative to stored procedures.
- Managed Supabase backups plus a documented restore test with measured recovery time.
- Sentry error monitoring, uptime alerting, structured JSON logging, and scheduled npm audit scanning in CI.
- bcrypt password hashing, scope-tagged JWTs with live status checks, Zod validation on all mutating routes, parameterised Drizzle queries, Helmet CSP, CORS allow-list, rate limiting, AES-256-GCM field encryption with a separate lookup hash, and row-locked transactions.
- Append-only financial ledger and `admin_audit_log` providing governance trails for money movement and back-office actions.
- Incident and postmortem log with a standard root-cause template, plus a technical risk register covering dependency vulnerabilities and single points of failure.
- Ongoing NIDA, BNR and NCSA regulatory compliance tracking as a standing control for a regulated product.

Production Software and AI Solutions (Click to View Live Demos)

KosmoInsight AI:

Live Demo - FinTech analytics platform with customer management, payment tracking, JWT authentication, CSV processing, demand forecasting, repayment risk prediction, customer segmentation and automated reporting. FastAPI, SQLAlchemy, PostgreSQL/SQLite, Next.js and TypeScript.

Kosmotive Ecosystem:

Live production platforms (KosmoPads, KosmoHealth, PayGo) built on Express, React and Supabase with MTN MoMo, Resend and SMS integrations, serving users across Rwanda, DRC, Central Africa and Kenya.

AB Rwanda Risk Analytics:

Live Demo - Full-stack risk analytics platform for microfinance. Real-time KPIs, risk alerts, fraud detection, stress testing, role-based dashboards and AI-assisted insights. Next.js, Supabase PostgreSQL, Claude API, Vercel.

RASIP, Rwanda Agri. Spatial Intelligence:

Live Demo | GitHub - Decision-support platform predicting crop suitability across 30 Rwanda districts. REST APIs, batch CSV analysis, SHAP explainability and report generation. FastAPI, Next.js 14, Supabase, TypeScript.

FarmSmart Agentic AI:

Live Demo - LLM advisory system using LangChain, RAG, tool-use agents and real-time API integrations. Deployed on Hugging Face Spaces.

Informal Address Resolver:

Live Demo - Low-latency geocoding service under 100ms, offline-capable, 94% accuracy within 300m.

Rwanda Air Quality Dashboard:

Live Demo - LSTM forecasting dashboard with health impact analysis. Presented at SOSCHI 2025.

Credit Risk Assessment:

GitHub - XGBoost with SHAP interpretability for explainable financial risk scoring.

Professional Experience

Kosmotive, Full-Stack Developer Fellow

2026 to Present

via AIMS-H2i Data Science Fellowship (Health Entrepreneurship, Africa Health Collaborative, University of Toronto)

- Build and operate the Kosmotive ecosystem (KosmoPads, KosmoHealth, PayGo) serving users across Rwanda, DRC, Central Africa and Kenya.
- Own the full lifecycle from requirements through design, development, integration, testing, deployment and 2nd/3rd-line support.
- Integrated MTN MoMo Collections API, Resend email and SMS gateway with signed webhooks, idempotent confirmation, retry queuing and interface documentation.
- Established CI/CD release gating with GitHub Actions and introduced automated test suites for the repayment and reconciliation engines.
- Designed the schema (18 tables, 29 foreign keys, 8 unique constraints) and added secondary indexes for admin and reporting queries.
- Implemented defence-in-depth security: bcrypt, scoped JWTs, Zod validation, parameterised queries, Helmet CSP, CORS, rate limiting, AES-256-GCM encryption and row locking.
- Delivered audit trails, governance records, incident logging, root-cause analysis, Sentry monitoring, uptime alerting, a tested backup-restore procedure and a technical risk register.

AIMS Research and Innovation Centre, Kigali

2023 to 2025

Software Solutions Engineer / ML Engineer and Data Manager

- Designed, built and maintained full-stack applications using Python (FastAPI), TypeScript/Next.js, PostgreSQL and Supabase.
- Implemented REST API integrations with third-party weather, geospatial and AI services with retries, error handling and monitoring.
- Established CI/CD pipelines and containerised deployments (Docker, GitHub Actions, Vercel, Render, Hugging Face Spaces).
- Developed unit, integration and regression tests; supported UAT and tracked defects through resolution.
- Provided 2nd/3rd-line support, root-cause analysis, proactive monitoring, patching and preventive maintenance.
- Led data integration for 120 IoT monitoring nodes including ETL pipelines, data quality assurance and PostgreSQL management.

Willows International, Kigali

2021 to 2022

Data Analyst / Reporting Engineer

- Built automated reporting pipelines, dashboards and data collection systems using Python, SQL and Power BI.
- Translated business requirements into technical designs and delivered decision-support tools for program evaluation.

Technical Skills

Software Engineering Node.js, Express Python (FastAPI) JavaScript, TypeScript React, Next.js HTML5, CSS3, Tailwind REST APIs and Web Services OOP and Design Patterns	Databases PostgreSQL, MySQL SQL Server, Oracle (familiar) Query Tuning and Indexing Schema Design and Migrations Drizzle ORM Backups and Restore Testing Data Integrity and Governance	Integration REST API Design Signed Webhooks MTN MoMo, Resend, SMS Data Mapping and Error Handling Retry Queues (Redis/BullMQ) Idempotency and Atomicity	QA and Testing Jest, Vitest Unit, Integration, Regression Test Automation in CI UAT and Defect Tracking Coverage Reporting
DevOps and Cloud Git and GitHub GitHub Actions CI/CD Docker and Linux/Bash Render, Vercel, Supabase AWS and GCP Basics	Security and Compliance OWASP Secure Coding bcrypt, Scoped JWT, RBAC Zod Validation, Helmet CSP AES-256-GCM Encryption BNR, NIDA, NCSA Compliance	Production Support Sentry Monitoring and Alerts Uptime Monitoring Structured JSON Logging Incident and Postmortem Logs Root-Cause Analysis Vulnerability Scanning	Data and AI Python (pandas, numpy) SQL, ETL/ELT, Power BI TensorFlow, PyTorch, XGBoost LangChain, RAG, Vector DBs SHAP, LIME, XAI

Core Competencies

Solution Delivery:	End-to-end design, development, integration, testing, deployment and maintenance of business-critical applications.
Full-Stack Development:	TypeScript/Node/Express and Python/FastAPI backends, React/Next.js frontends, REST APIs and reporting dashboards.
Systems Integration:	Payment, email and SMS integration; signed webhooks; idempotency; retry queues; interface documentation.
Database Engineering:	Normalised schema design, query tuning, secondary indexing, versioned migrations, encryption at rest, tested restore procedures.
Quality Engineering:	Test strategy, automated suites in CI, UAT coordination and full defect lifecycle traceability.
Production Support:	2nd/3rd-line support, incident logging, root-cause analysis, proactive monitoring, alerting and vulnerability scanning.
Security and Controls:	OWASP-aligned secure coding, access control, encryption, audit trails and support for internal and external IT audits.
Financial and Health Systems:	Payments, ledgers, reconciliation, credit risk scoring, fraud detection and digital health platforms.

Education

African Institute for Mathematical Sciences (AIMS), Rwanda	2021 to 2023
<i>MSc in Mathematical Sciences (Major: Data Science)</i> , Graduated June 2023	
Thesis: Machine Learning and Time Series Analysis for Environmental Early Warning Systems [View]	
University of Rwanda	2018 to 2021
<i>BSc in Statistics</i> . Coursework: Biostatistics, Survival Analysis, Econometrics, Design of Experiments, Time Series Forecasting.	

Selected Certifications and Training

- AIMS RIC Doctoral Training School 2026 - Foundational Methods in Data Science [\[Certificate\]](#)
- East-African School on Advanced Machine Learning (EASY-ML) - CNRS, Sorbonne, University of Nairobi (2025)
- Modern Statistics and Machine Learning for Global Health - Imperial College London and AIMS (2024)
- Synthetic Data for Healthcare Research - Washington University St. Louis, DS-I Africa, AIMS Rwanda, NIH (2023)
- Building Your Own Chatbot: RAG, Fine-tuning LLMs, Deployment - AIMS RIC (2025)
- Geospatial Analytics using R and R-INLA - MRC, Imperial College (2023)

- ICDL Data Analytics Certification - ICDL Africa (2024)
- Certificate in Business Management - ESMT Berlin, iiAfrica, Harvard Business Publishing (2023)
- App Development with Swift - Apple Authorised Training (2025)
- Monitoring and Evaluation for Development Programs - World Bank Group (2024)

[All certificates and degrees: [Click to view](#)]

Research Publications and Recognition

- **Co-author:** *City-wide space-time patterns of environmental noise pollution in Kigali, Rwanda*. Environmental Research Letters, 2024 [\[View Paper\]](#)
- **Presenter:** *Early Warning System for Air Pollution in Rwanda*. SOSCHI Conference on Standards for Official Statistics on Climate-Health Interactions, Kigali, December 2025 [\[Certificate\]](#)
- **Certificate of Recognition:** MLOA 2025, Second African Conference on Machine Learning, Optimization and Applications, AIMS and TU Ilmenau.
- Full-Stack Developer Fellow - Kosmotive, via AIMS-H2i Data Science Fellowship, HENT and Africa Health Collaborative, University of Toronto (2026).
- AIMS KTT Hackathon Participant - Data Engineering, Agentic AI and Geospatial Tracks (2025 to 2026).

References

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